

Pratik Bairoliya

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EXPERIENCE

Capital One

Jul. 2026 – Present

Software Engineer II

McLean, VA

- Own the nightly **end-of-day close** — posting, settlement, ledger cutoff across Card and Bank, run on **Temporal**.
- Lead the **REST API** replacing the vendor's close operator tooling, bringing nightly controls in-house.
- Monitoring and distributed tracing in **OpenTelemetry** and **Splunk**: alerts page on-call mid-run, not at cutoff.
- Ship end-of-day services on **Kubernetes** through **Argo CD** and **Helm**, rollback one revision away.

Capital One

May 2024 – Jul. 2026

Software Engineer I

McLean, VA

- Wrote the ledger's **batch write path** in **Go**, empty repo to production; every nightly cycle runs through it.
- Designed a **block-allocation ID service**: one call reserves a range, taking per-ID contention off bulk opens.
- Built the nightly **PySpark/Databricks** reconciliation: day-over-day aggregate diffs alert before a bad cycle ships.
- Cut data egress on the nightly **99M+ Card account** publish by performance-tuning **Hive** sort on **AWS EMR**.
- Rolled **proof-of-possession (PoP) tokens** across Card and Bank endpoints, binding each call to a key a thief lacks.
- Rebuilt the military rewards catalog in **Vue.js/Spring Boot/DynamoDB**: **2.8x** faster, \$9M projected revenue.

Fidelity Investments

Jun. 2023 – May 2024

Software Engineer

Durham, NC

- Moved mainframe stock and job workflows to **AWS** with **Terraform** — **45%** of volume, **400+** associates.
- Modeled system metadata as a **Neo4j** graph; **Cypher** queries found and removed **5,000+** duplicate records.
- Delivered the full stack on a 4-person team: **Angular** UI, **Java** services, **S3** metadata sync.

PROJECTS

Lookback — automated daily briefing | *Python, Ollama (local LLM), Google APIs, SQLite, launchd* 2026

- Scheduled job merging Calendar, Gmail, and iMessage into one morning note; runs **100% on-device**.
- **Safe to re-run**: a missed night rebuilds the same note instead of duplicating it.

Pantry — receipt-to-groceries tracker | *Python, Apple Vision OCR, Ollama, file watchers* 2026

- **OCR** ingest pipeline turning receipt photos into a pantry list with spend rolled up by category.
- **Idempotent ingest**: rescanning a receipt cannot double-count, so spend totals stay correct.

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript/JavaScript, SQL, Cypher, Bash, Git

Cloud & Infrastructure: AWS (EMR, S3, DynamoDB), Linux, Kubernetes, containerization (Docker), workflow orchestration (Temporal), Terraform (IaC), Argo CD, Helm, GitOps, Jenkins, CI/CD

Backend & Data: REST API design, microservices, distributed batch processing, ETL, Apache Spark (PySpark), Hive, Databricks, idempotency, retries, authentication (PoP tokens), Spring Boot, Neo4j

Also: Vue.js, Angular, TypeScript, HTML/CSS, local LLM inference (Ollama), OCR pipelines

Practices: Distributed systems design, observability (Splunk, OpenTelemetry), on-call incident response

Certification: AWS Certified Solutions Architect – Associate (Mar. 2026)

EDUCATION

North Carolina State University

Aug. 2021 – May 2025

B.S. Computer Science and B.S. Applied Mathematics — GPA 4.0

Raleigh, NC

- **Park Scholar** — full merit scholarship to the top **1%** of applicants for leadership and service.
- **Teaching Assistant** — CSC 221 (Python) and MA 405 (Linear Algebra): labs, office hours, grading.
- **Undergraduate Researcher**, 2022–2024 — built **Python** tooling that automated a faculty group's data analysis.
- **Chairman, InspireNC** — student-run nonprofit teaching FIRST Robotics in NC schools; captain of Team 6908.

INTERESTS

Underwater robotics, running the Krispy Kreme Challenge, and home-lab automation.